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Implementation of circularity in food supply chain based on big data techniques using Einstein's fuzzy methods

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By emphasizing the principles of recycling, refurbishing, and reusing materials, circularity in supply chain management plays a pivotal role in waste reduction and resource efficiency enhancement. However, the adoption and implementation of circular supply chains (CSCs) remain limited due to the substantial associated risks. Research in Circular Supply Chain Management (CSCM) suggests that integrating Fourth Industrial Revolution technologies and big data analytics may offer a viable solution. Despite growing interest in CSCM, the lack of field-specific data and a predominant focus on macro-level studies have resulted in insufficient exploration of circularity barriers within Iran's food supply chain. This study seeks to address this gap by prioritizing big data-driven solutions to overcome these barriers. The research objectives include identifying and validating key obstacles within Iran's food industry through an extensive literature review and content validity ratio analysis. The barriers were classified into five primary categories, with their respective weights determined using the fuzzy SWARA (Step-wise Weight Assessment Ratio Analysis) method. The most critical barriers identified were the absence of organizational infrastructure to support circular supply chain initiatives and the lack of essential systems for product traceability. Subsequently, big data-based techniques were evaluated and prioritized using Einstein's fuzzy WASPAS (Weighted Aggregated Sum Product Assessment) method. Combining the robustness of fuzzy logic with MCDM (Multi-Criteria Decision-Making) techniques, this approach offers an effective framework for weighting circular supply chain barriers and prioritizing big data techniques under conditions of uncertainty. Notably, this study represents the first application of Einstein's fuzzy WASPAS method in circular supply chain management. The findings reveal that SNA (social network analysis) and optimization models exert the most significant influence in mitigating implementation barriers. These insights can guide food industry managers in strategically adopting smart technologies to address key challenges in circular supply chain adoption.

Keywords Circularity, Circular supply chain, Food industry, Big data, Einstein's fuzzy

In the context of the rapidly expanding global economy, excessive consumption and production have markedly exacerbated environmental degradation, including climate change, global warming, resource scarcity, and ecological decline [49]. Both consumers, through unsustainable consumption behaviors, and producers, via conventional resource-intensive production methods, have substantially contributed to these pressing social and environmental challenges. Given that supply chains serve as the critical link between consumers and producers, sustainability has been widely advocated as a key solution to mitigate environmental and social harm. This concept has consequently gained significant attention in supply chain research [74]. Among emerging sustainability paradigms, the circular economy (CE) has been recognized as a more ecologically viable alternative to the traditional linear economic model [14, 19]. By integrating circular economy principles into supply chain management, this approach introduces an innovative framework for sustainable operations, prompting a surge in scholarly interest [14].

The fundamental aim of CE is to preserve the utility of products and materials by prolonging their lifecycle through circular design, thereby maximizing value creation, minimizing waste generation, and optimizing natural resource utilization [4, 26]. A core tenet of this model is that waste from one entity can function as a valuable input for another [64], presenting substantial opportunities for improving organizational sustainability

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performance. As a result, the incorporation of circular economy principles into supply chain management (SCM) has attracted growing research attention. Nevertheless, considerable ambiguity persists regarding the terminology and conceptual boundaries of sustainable supply chain practices [19, 21].

The circular supply chain constitutes an advanced management paradigm that promotes the utilization of renewable, recyclable, or biodegradable materials throughout production processes [36]. The implementation of CE principles within supply chain operations serves to substantially minimize waste generation [45] while concurrently mitigating environmental impacts [11]. As such, the adoption of CSC models presents significant potential for reducing ecological burdens, particularly within the food industry where waste and material loss reduction are of paramount importance. A critical distinction emerges when examining food systems in contrast to general production systems. Food supply chains predominantly operate on linear models. This structural characteristic contributes to substantial inefficiencies, with approximately 1.3 billion tons—representing nearly one-third of total food production for human consumption—being lost during harvest, processing, and distribution phases, or ultimately wasted at the retail and consumption stages [30]. The inherent properties of food products, including perishability [77], specialized handling requirements, and stringent storage conditions, present unique challenges within supply chain management. The Iranian food industry exemplifies these complexities, comprising multiple interdependent stages that collectively exert significant influence on public health outcomes [39]. Among the most pressing operational challenges confronting this sector are escalating waste volumes and deteriorating quality of perishable goods [52, 58]. Addressing these challenges necessitates strategic investments in comprehensive corrective measures encompassing process optimization, technological upgrades, and infrastructure improvements. In this context, the implementation of circular supply chain methodologies may offer an effective solution for enhancing product quality while simultaneously reducing systemic waste.

However, the implementation of circular practices in developing nations such as Iran presents greater challenges compared to developed countries, owing to disparities in regulatory frameworks, infrastructural development, and socio-cultural conditions. Commercial and industrial enterprises typically face substantial capital expenditures when adopting CSC models, necessitating active collaboration from suppliers and retailers, as successful implementation requires participation from all value chain stakeholders [16]. Within this framework, both suppliers and manufacturers encounter multiple impediments to CSC adoption, including limited conceptual understanding, financial constraints, and managerial limitations. Consequently, the systematic identification and analysis of barriers to CSC implementation represents a critical research imperative that warrants comprehensive examination. The enablers and barriers to CSC adoption demonstrate significant variation across industrial sectors and operational contexts. This heterogeneity underscores the necessity for context-specific research to prioritize influential factors and develop targeted intervention strategies for barrier mitigation. Technological innovation and digital transformation have fundamentally reshaped contemporary business operations, including supply chain management paradigms. These developments have enhanced organizational efficiency across production, consumption, and operational processes. Big Data (BD) analytics has emerged as a particularly transformative tool in SCM, enabling enterprises to: (1) collect precise product information, (2) facilitate product traceability, (3) monitor distribution networks, and (4) optimize delivery systems [43]. Supply chain-derived data now serves as a critical resource for industrial analysis, pattern recognition, and strategic decision-making. The contemporary business environment has transitioned from physical data storage systems to advanced big data infrastructures capable of processing exponentially increasing data volumes [70, 72]. Industries increasingly rely on data analytics to mitigate supply chain risks [72], while enterprises utilize these insights to optimize operational processes and enhance overall supply chain performance through evidence-based solutions.

In light of the considerable importance and practical applicability of big data-driven solutions, this research systematically identifies and prioritizes such approaches to mitigate barriers impeding the implementation of CSCs within the food industry. Building upon the previously established significance of CSC adoption, this study aims to develop a comprehensive framework demonstrating the efficacy of big data-based solutions in addressing these implementation challenges. Given the methodological constraints associated with examining all industrial sectors in Iran, the research scope has been deliberately circumscribed to Fars province and its representative industries.

Notwithstanding the increasing significance of CSC implementation in the food industry, scholarly investigation within the Iranian context remains limited, particularly with respect to the systematic identification and classification of implementation barriers. Existing literature predominantly examines generic challenges including infrastructural inadequacies, economic constraints, and policy support deficiencies, while largely neglecting context-specific cultural, managerial, and technological barriers unique to Iran. For illustration, recent research conducted in Kurdistan Province [31] delineated three primary impediments to CE adoption in agricultural supply chains: inadequate governmental support, absence of sustainable production methodologies, and insufficient producer incentives. Given the distinctive structural and operational characteristics of Iran's food sector, this study proposes a bespoke analytical framework specifically designed to identify and prioritize implementation barriers with greater precision and contextual relevance. For data analysis, this investigation employs an integrated methodological approach combining fuzzy SWARA and fuzzy Einstein WASPAS. This hybrid methodology yields novel insights by virtue of its superior capacity to process the inherent complexities and uncertainties characteristic of supply chain decision-making processes. Conventional MCDM approaches, which predominantly rely on crisp data, prove inadequate for addressing the fuzzy and volatile information environments typical of real-world supply chain systems. Such traditional methods demonstrate limited effectiveness in managing uncertainty and complexity. Conversely, the integrated fuzzy SWARA and Einstein's fuzzy WASPAS approach offers distinct advantages:

- Enhanced analytical precision through fuzzy data processing, enabling better alignment with dynamic supply chain conditions;
- Improved capacity for handling complex, MCDM scenarios;
- Superior modeling of non-linear inter-criteria relationships through fuzzy SWARA, permitting more comprehensive option evaluation; and
- Advanced computational capabilities via the novel integration of WASPAS with Einstein's theory, particularly effective in contexts characterized by intricate criterion dependencies.

The synergistic application of these methodologies consequently produces more robust and optimal outcomes in complex, uncertain operational environments.

The rest of the article is structured as follows. The following section presents the literature review and background of the study. The third section describes the proposed methods. The fourth section presents the results of the analysis. The fifth section presents a discussion of the results, and the sixth section examines the conclusions and implications of the study.

Literature review

Circular supply chain

The conventional supply chain paradigm adheres to a "take-make-dispose" linear model, which proves fundamentally inadequate in maintaining equilibrium between supply–demand networks and the utilization of natural resources including raw materials, energy, and fossil fuels [27, 50]. This systemic imbalance in production and consumption patterns exerts substantial detrimental effects on ecosystem sustainability [9]. Consequently, in recognition of impending resource scarcity challenges, industrial sectors must urgently transition from linear to CSC models. The CE paradigm offers an innovative framework for transforming conventional consumption systems into regenerative cycles [73]. Comparative analysis reveals distinct supply chain typologies: linear models extract and dispose of materials throughout product lifecycles, while closed-loop systems enhance environmental performance through product and packaging recovery [28]. Despite its benefits, a closed-loop supply chain still generates significant amounts of waste due to the lack of reuse or recycling opportunities for all unwanted materials along the supply chain. CSC represent an advanced paradigm designed for zero-waste output through systematic resource regeneration. These systems maintain two distinct material flows: primary (linear/closed-loop) and circular (re-flow) pathways, the latter encompassing recycling, maintenance, reuse, repair, and restoration operations [28]. Recent systematic reviews [82] identify five predominant research clusters in CSC literature: (1) reverse channel optimization, (2) reviews of CSCM and empirical studies, (3) closed-loop supply chains and consumers, (4) closed-loop supply chains and inventory management, and (5) closed-loop supply chains and reverse logistics. Montag [60] underscores the dynamic and rapidly evolving nature of CSC research, noting persistent conceptual ambiguities in its differentiation from adjacent sustainability domains. Current literature demonstrates significant variation in the definitional parameters, conceptual scope, and thematic focus of CSCM. This conceptual fragmentation is further substantiated by Dey et al. [14], who identify the absence of a comprehensive theoretical framework for CSCM as a critical impediment to scholarly advancement in the field. To address these limitations, future research should prioritize the development of robust performance metrics, systematic identification of implementation barriers, formulation of novel circular business models, and design of innovative structural frameworks. Hazen et al. [34] contribute to this discourse through their analytical integration of five fundamental CE principles (material loop closure, resource intensity reduction, product life extension, cycle intensification, and dematerialization) with eight core supply chain processes. Their work elucidates specific mechanisms through which conventional supply chain operations can facilitate the transition from linear to circular production-consumption systems.

The foundational work of Farooque et al. [19, 21] represents a seminal contribution to the field, establishing the first comprehensive taxonomy of supply chain sustainability terminology and proposing a unified definition of CSCM. Through systematic analysis of 261 scholarly articles, their structured literature review not only maps the current research landscape but also provides actionable guidelines for implementing circular economy principles across supply chain echelons. Notably, their work identifies several emergent research domains including circular and regenerative design paradigms, remanufacturing processes, extended product and producer responsibility frameworks, and the integration of advanced digital technologies in CSCM implementations.

Circular supply chain barriers

Existing scholarly work has extensively documented the multifaceted challenges inherent in adopting CSC models across various industrial contexts. This necessitates the development of robust, organization-centric solutions to address these implementation barriers. A critical examination reveals that CSCM adoption challenges exhibit significant industry-specific variations, requiring both precise identification and scalable solution frameworks [24, 25, 27, 62]. This industry-specific nature of implementation barriers underscores the importance of sectoral analysis in CSC research. The extant literature from the past decade demonstrates growing academic interest in CSC models and their associated implementation challenges. Recent empirical studies have yielded important sector-specific findings. Mahmoudi et al. [53] identified lack of effective management as the primary implementation barrier within Iran's copper industry. Lahane and Kant's [49] investigation of Indian manufacturing revealed the lack of top management support as the most critical adoption challenge. Farooque et al. [20] established that strategically implemented CSCM practices significantly enhance cost efficiency and financial performance in Chinese industries. Khan and Ali [40] explored the implementation of CSCs in the pharmaceutical industry. They highlighted three predominant challenges including financial constraints, market-related obstacles, and supply chain coordination deficiencies. Kazancoglu et al. [38] developed a dairy industry framework emphasizing economic barriers as most consequential, proposing big data analytics as potential solutions. Lahane and Kant [48] demonstrated low CSC adoption rates with perceived implementation risks, developing mitigation strategies through solution prioritization. The purpose of their research is to identify and rank solutions to mitigate the risks in implementing CSCs. Khandelwal et al. [41] present research on the implementation of CSCM in the plastics industry in India. The results indicate that the lack of tax relief policies and the weak enforcement of environmental protection laws and regulations are the most significant barriers. Farooque et al. [19, 21] established two fundamental barriers in China's food supply chains including insufficient environmental legislation and the lack of market-driven pressures. Govindan et al. [26] provided an analysis of the drivers, barriers, and practices affecting the implementation of the CE in the context of supply chains through a systematic review. The overview of predominant CSCM models and their objectives presents in Online Appendix (A).

Big data techniques

The conceptualization of Big Data remains contested across academic, industrial, and professional domains, with no universally accepted definition currently established. This definitional plurality contributes to significant conceptual ambiguity regarding the precise nature and boundaries of big data [79]. Operational definitions typically demonstrate sector-specific variations, being contingent upon both available analytical tools and the characteristic data volumes within particular industries [7]. From a technical perspective, Big Data constitutes extensive datasets characterized by high degrees of structural heterogeneity and organizational complexity. These datasets surpass the processing capacity of conventional database architectures and computational infrastructures [1, 66]. Within the present research context, Big Data analytics refers to the systematic examination of voluminous datasets to identify latent patterns, correlations, and insights. Such analytical outputs enable organizational actors to enhance situational awareness and capitalize on emergent opportunities.

The advent of Industry 4.0 technologies and smart production systems has precipitated the development of multiple Big Data frameworks for CSCM. Gammelgaard and Nowicka [23] posit cloud-based internet platforms and digital ecosystems as critical information nexuses for supply chain stakeholders. Faisal's [18] empirical investigation elucidates Industry 4.0's mediating role between CSCM drivers and circular performance outcomes, establishing significant positive relationships between these constructs. Choi and Chen [10] present pioneering work demonstrating the application of Big Data analytics to large-scale group decision-making processes in CSCM contexts. Complementarily, Del Giudice et al. [12] examine the performance implications of CE practices within supply chains, with particular emphasis on the moderating effects of Big Data-enabled supply chain architectures. Big data analytics can be systematically classified into seven principal methodological domains: optimization techniques, data mining approaches, machine learning algorithms, cloud computing architectures, statistical methodologies, artificial neural network systems, and social network analysis frameworks [35]. ***Optimization Models*** incorporate advanced computational techniques including linear programming models, nonlinear programming solutions, evolutionary computation algorithms, and graph-based optimization approaches. These methodologies significantly enhance operational efficiency by improving production throughput, resolving distribution network challenges, and optimizing inventory control systems [8, 37]. ***Data Mining*** processes employ a comprehensive suite of analytical techniques to extract meaningful patterns, correlations, and actionable insights from voluminous datasets. This domain utilizes sophisticated analytical tools including cluster analysis, classification algorithms, regression models, association rule mining, and temporal pattern recognition systems [32]. ***Machine Learning*** represents a fundamental component of big data analytics, enabling the automated discovery of complex data structures and the generation of predictive models from extensive datasets [75]. ***Cloud Computing*** Infrastructures facilitate ubiquitous access for all supply chain stakeholders, significantly enhancing information transparency and mitigating knowledge transfer inefficiencies across organizational boundaries [3, 17, 63]. This technological paradigm fosters the development of CSC ecosystems by enabling knowledge dissemination and workforce capacity building for CE implementation [37, 71]. ***Statistical Techniques*** constitute essential methodologies for data exploration and knowledge extraction, encompassing multivariate regression analysis, inferential hypothesis testing, unsupervised clustering, supervised classification, and dimensionality reduction through factor analysis [61]. These techniques enable effective management of large-scale datasets and the extraction of meaningful patterns and insights. ***Artificial Neural Network*** consists of interconnected processing units with adaptive connection weights. These systems are particularly effective in modeling the complex, nonlinear relationships between supply chain entities and processes, thereby addressing the dynamic and multifaceted nature of modern supply chains [80]. ***Social Network Analysis*** applies graph theory principles to examine structural relationships and behavioral patterns within interconnected systems. This analytical approach, frequently implemented through advanced data mining techniques, has demonstrated particular efficacy in processing big data volumes that exceed conventional analytical capabilities. It serves as a critical decision-support tool for supply chain managers in policy formulation and strategic planning [37, 42].

Methodology

The objective of this study is to identify and prioritize Big Data techniques capable of addressing the barriers hindering the implementation of CSCs in the food industry. The findings hold significant practical value for industry managers handling perishable and recyclable products in the market. To achieve this objective, a systematic literature review on CSCs was conducted, employing the PRISMA methodology for analysis. To validate and prioritize the identified barriers, expert opinions were collected from 30 professionals across 15 small and medium-sized enterprises (SMEs) in Iran's food industry. The participating companies operate in diverse sectors, including dried fruit, dairy, meat products, ready-to-cook foods, ready meals, protein products, and canned/pickled cucumber production. The respondents held key positions in marketing, research & development (R&D), and production/operations management. Additionally, two academic experts have served as consultants to dairy industry firms for over five years, providing specialized insights into this sector. Demographic information related to research experts have been mentioned in Online Appendix (B).

This research was structured into three key phases. In the first stage, following an extensive literature review, the barriers to CSC implementation and corresponding Big Data-driven solutions were identified, refined, and validated through expert consensus. In the second stage, the relative importance of each barrier was determined using the Fuzzy SWARA method. In the third stage, Big Data-based solutions (alternatives) were ranked using the Fuzzy Einstein WASPAS method. Expert input was leveraged at critical stages: 10 experts contributed to the Fuzzy SWARA analysis, while 7 participated in the Fuzzy Einstein WASPAS evaluation. Figure 1 illustrates the data analysis framework.

Content validity ratio (CVR)

To assess and refine the barriers to circular supply chain (CSC) implementation in the food industries, the content validity ratio (CVR) method, as introduced by Lawshe [51], was employed. The CVR for each barrier was computed using Eq. (1), incorporating expert evaluations, where *N* represents the total number of participating experts and n_e denotes the subset of experts who classified the barrier as "highly critical". The minimum acceptable CVR threshold was determined based on the total number of expert assessors.

$$\mathrm{CVR} = \frac{ne - \frac{N}{2}}{\frac{N}{2}} \tag{1}$$

Fuzzy SWARA

This study employs the fuzzy SWARA method, originally developed by Mavi et al. [57], to determine criterion weights. The selection of this methodology is justified by its distinct advantages in addressing MCDM problems under conditions of uncertainty. In comparison to alternative fuzzy MCDM techniques such as fuzzy Analytic Hierarchy Process (FAHP) or fuzzy Analytic Network Process (FANP), the fuzzy SWARA method offers a more streamlined approach that significantly reduces expert cognitive burden. This is achieved through its requirement of sequential judgments rather than exhaustive pairwise comparisons. The method's step-wise architecture, incorporating adjustment coefficients, facilitates a systematic and adaptable elicitation of expert judgments—a particularly advantageous feature in decision environments characterized by ambiguity or imprecision. Moreover, the fuzzy SWARA method demonstrates superior practicality when compared to more resource-intensive approaches like fuzzy ANP. While maintaining robust capability in capturing subjective preferences, it demands substantially fewer inputs, rendering it particularly suitable for decision contexts constrained by limited data availability, resource scarcity, or human-related uncertainties. The implementation procedure for this method follows a well-defined six-step process, as comprehensively described by Perçin [65].

Step 1: The criteria should be classified according to their importance.

In this step, experts rank the defined barriers (criteria) according to their importance. The most important value is in the first rank, the least value is in the last rank, and other values are placed in the middle range of these two based on their importance.

Step 2: Determining the comparative importance of criteria.

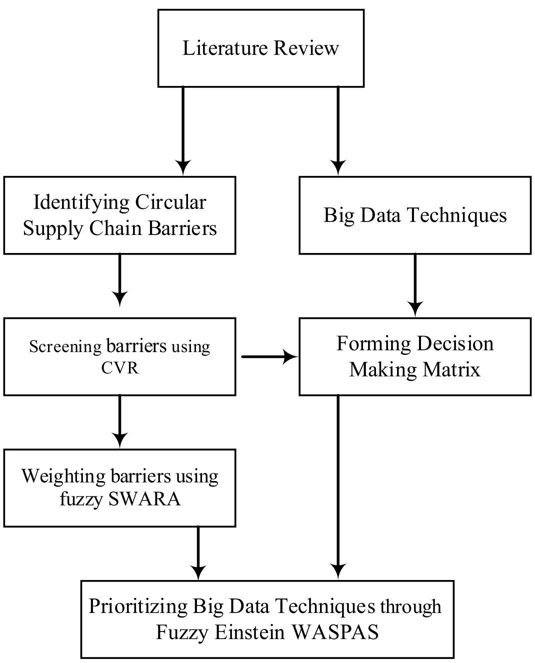

Fig. 1. Research process.

Starting from the second-ranked criterion, each expert evaluates to what extent each criterion (*cj + 1*) is less important compared to the previous one (*cj*), based on their subjective judgment and linguistics terms mentioned in Table 1. This relative importance is expressed as a coefficient that reflects the perceived decrease in significance.
Step 3: We calculate the kj coefficient as follows.

$$K_j = \begin{cases} 1, & j = 1 \\ s_{j+1}, & j > 1 \end{cases} \tag{2}$$

Step 4: Measure the recalculated weight (qj) as follows.

$$q_j = \begin{cases} 1, & j = 1 \\ \frac{q_{j-1}}{k_j}, & j > 1 \end{cases} \tag{3}$$

Step 5: Calculating the weights of the criteria so that their sum is equal to one.

$$W_j = \frac{q_j}{\sum_{k=1}^{m} q_j} \tag{4}$$

Step 6: We convert the weights related to fuzzy relative importance to the crisp numbers based on the surface center method using the following relationship.

$$w_j^{non} = \frac{\left(w_j^u - w_j^l\right) + \left(w_j^m - w_j^l\right) + \left(w_j^l\right)}{3} \tag{5}$$

A number with a higher value will have a higher rank [65].

Einstein fuzzy WASPAS

The WASPAS method, which is one of the multi-criteria decision-making techniques, was developed by Zavadskas et al. [81]. WASPAS includes a weighted sum model (WSM) and a weighted product model (WPM) to provide decision making. In this study, the classic WASPAS method is improved by integrating fuzzy Einstein operators in a fuzzy environment on triangular fuzzy numbers. By integrating Einstein's functions in the model, it helps to objectify decision-making. The advantages of employing fuzzy Einstein operations in MCDM methods such as WASPAS can be summarized as follows:

1. The fuzzy Einstein WASPAS model improves the management of group-based data and effectively addresses uncertainties inherent in expert judgments.
2. The proposed MCDM framework integrates Einstein norms, thereby enhancing the efficacy of the conventional WASPAS approach [81].
3. Within the WASPAS model, Einstein norms are utilized to transform linear weighted sum and weighted product functions into nonlinear formulations.
4. Traditional weighted sum and product operations are substituted with fuzzy Einstein weighted averaging and geometric averaging functions, which facilitate nonlinear data processing. These modifications further enhance the model's flexibility in dynamic decision-making environments.

We assume that $O_i = (O_1, O_2, \ldots O_3)(i = 1,2, \ldots, n)$ are the solutions, $K_i = (K_1, K_2, \ldots K_m)$ $(j = 1,2, \ldots, m)$ is the criteria set and $(E_1, E_2, \ldots E_e)(l = 1,2, \ldots, e)$ and a set of experts and weight coefficients of criteria $w_j = (w_1, w_2, \ldots, w_m)^T$. The stages of the proposed WASPAS Einstein Fuzzy model are as follows [13]:
Step 1. Aggregated decision matrix $D = [d_{ij}]_{n \times m}$ is obtained by using individual decision matrices based on the linguistic terms depicted in Table 2.
Step 2. The decision matrix $D = [d_{ij}]_{n \times m}$ is normalized using the Eq. (6). According to the type of benefit (B) and cost (C) criteria.

Linguistic terms	Fuzzy numbers
Very high (VH)	(4,5,5)
High (H)	(3,4,5)
Medium (M)	(2,3,4)
Low (L)	(1,2,3)
Very low (VL)	(1,1,2)

Table 1. Linguistic terms [65].

Linguistic terms	Fuzzy numbers
Absolutely Low (AL)	(1,1,1)
Very Low (VL)	(1,2,3)
Low (L)	(2,3,4)
Medium low (ML)	(3,4,5)
Equal (E)	(4,5,6)
Medium high (MH)	(5,6,7)
High (H)	(6,7,8)
Very high (VH)	(7,8,9)
Absolutely High (AL)	(8,9,9)

Table 2. Linguistic terms for evaluating alternatives [13].

$$= \begin{cases} X_{ij}^{(a)} = \frac{d_{ij}^{(a)}}{d_j^{+}}; X_{ij}^{(b)} = \frac{d_{ij}^{(b)}}{d_j^{+}}; X_{ij}^{(a)} = \frac{d_{ij}^{(c)}}{d_j^{+}} if j \in B, \\ X_{ij}^{(a)} = \frac{d_j^{(-)}}{d_{ij}^{(a)}}; X_{ij}^{(b)} = \frac{d_j^{(-)}}{d_{ij}^{(b)}}; X_{ij}^{(a)} = \frac{d_j^{(-)}}{d_{ij}^{(c)}} if j \in C. \end{cases} \tag{6}$$

$\mathrm{X}_{ij}^{s} = (\mathrm{X}_{ij}^{(a)}, \mathrm{X}_{ij}^{(b)}, \mathrm{X}_{ij}^{(c)})$ represents the normalized matrix $\mho = [X_{ij}]_{n\times m}$ The d_j^{+} and d_j^{-} elements are determined using the Eqs. (7) and (8).

$$d_j^{+} = \max_{1\leq j\leq m} (\mathrm{d}_{ij}^{(a)}, \mathrm{d}_{ij}^{(b)}, \mathrm{d}_{ij}^{(c)}) \tag{7}$$

$$d_j^{-} = \min_{1\leq j\leq m} (\mathrm{d}_{ij}^{(a)}, \mathrm{d}_{ij}^{(b)}, \mathrm{d}_{ij}^{(c)}) \tag{8}$$

The Fuzzy Einstein WASPAS model includes two weighted sequences to calculate alternative aggregation strategies. The first weighted sequence is expressed using the fuzzy Einstein weighted mean function, while the second weighted sequence is expressed using the fuzzy Einstein geometric weighted mean function as follows:

Let $(\widetilde{X}_1, \widetilde{X}_2, \ldots, \widetilde{X}_m)$ be a set of normalized matrix elements $\mho = [X_{ij}]_{n\times m}$ with fuzzy numbers $\widetilde{X}_{ij} = (\mathrm{X}_{ij}^{(a)}, \mathrm{X}_{ij}^{(b)}, \mathrm{X}_{ij}^{(c)})$ is shown where $(j = 1{,}2, \ldots, m; i = 1{,}2, \ldots, n)$ the fuzzy Einstein weighted average function can be recognized as Eq. (9):

$$WS_i = (\mathrm{WS}_i^{(a)}, \mathrm{WS}_i^{(b)}, \mathrm{WS}_i^{(c)}) = \begin{cases} \sum_{j=1}^{m} (X_{ij}^{(a)}) \frac{\prod_{j=1}^{m} (1+f(X_{ij}^{(a)}))^{1/e} - \prod_{j=1}^{m} (1-f(X_{ij}^{(a)}))^{1/e}}{\prod_{j=1}^{m} (1+f(X_{ij}^{(a)}))^{1/e} + \prod_{j=1}^{m} (1-f(X_{ij}^{(a)}))^{1/e}} \\ \sum_{j=1}^{m} (X_{ij}^{(b)}) \frac{\prod_{j=1}^{m} (1+f(X_{ij}^{(b)}))^{1/e} - \prod_{j=1}^{m} (1-f(X_{ij}^{(b)}))^{1/e}}{\prod_{j=1}^{m} (1+f(X_{ij}^{(b)}))^{1/e} + \prod_{j=1}^{m} (1-f(X_{ij}^{(b)}))^{1/e}} \\ \sum_{j=1}^{m} (X_{ij}^{(c)}) \frac{\prod_{j=1}^{m} (1+f(X_{ij}^{(c)}))^{1/e} - \prod_{j=1}^{m} (1-f(X_{ij}^{(c)}))^{1/e}}{\prod_{j=1}^{m} (1+f(X_{ij}^{(c)}))^{1/e} + \prod_{j=1}^{m} (1-f(X_{ij}^{(c)}))^{1/e}} \end{cases} \tag{9}$$

where $w_j = (w_1, w_2, \ldots, w_m)^T$ represents the fuzzy weight coefficients of the criteria, while:

$$f(\tilde{X}_j) = \left(X_j^{(a)} \Big/ \sum_{j=1}^{m} X_j^{(a)}, X_j^{(b)} \Big/ \sum_{j=1}^{m} X_j^{(b)}, X_j^{(b)} \Big/ \sum_{j=1}^{m} X_j^{(b)} \right) \tag{10}$$

$$WP_i = (\mathrm{WP}_i^{(a)}, \mathrm{WP}_i^{(b)}, \mathrm{WP}_i^{(c)}) = \begin{cases} \sum_{j=1}^{m} (X_{ij}^{(a)}) \frac{2\prod_{j=1}^{m} (1+f(X_{ij}^{(a)}))^{w_j^{(a)}}}{\prod_{j=1}^{m} (2-f(X_{ij}^{(a)}))^{w_j^{(a)}} + \prod_{j=1}^{m} (f(X_{ij}^{(a)}))^{w_j^{(a)}}} \\ \sum_{j=1}^{m} (X_{ij}^{(b)}) \frac{2\prod_{j=1}^{m} (1+f(X_{ij}^{(b)}))^{w_j^{(b)}}}{\prod_{j=1}^{m} (2-f(X_{ij}^{(b)}))^{w_j^{(b)}} + \prod_{j=1}^{m} (f(X_{ij}^{(b)}))^{w_j^{(b)}}} \\ \sum_{j=1}^{m} (X_{ij}^{(c)}) \frac{2\prod_{j=1}^{m} (1+f(X_{ij}^{(c)}))^{w_j^{(c)}}}{\prod_{j=1}^{m} (2-f(X_{ij}^{(c)}))^{w_j^{(c)}} + \prod_{j=1}^{m} (f(X_{ij}^{(c)}))^{w_j^{(c)}}} \end{cases} \tag{11}$$

Step 4. The weighted aggregation of the fuzzy Einstein weighted average function and the fuzzy Einstein geometric weighted average function is calculated as Eq. (12):

$$\Delta_i = \lambda . WS_i + (1-\lambda) WP_i \tag{12}$$

λ is the parameter of the thank you method and the range is 0–1. When the value of λ is 1, WASPAS leads to WPM, while for λ = 1, WASPAS becomes WSM.

Step 5. Δ_i is defuzzified by the Eq. (13).

$$D_i = \frac{\Delta_i^{(a)} + 4\Delta_i^{(b)} + \Delta_i^{(c)}}{6} \tag{13}$$

Step 6. Options are ranked in descending order according to D_i values.

Results

To accomplish the research objectives, the initial phase involved systematic identification of barriers hindering CSC implementation. A comprehensive literature review was conducted across multiple academic databases, including ScienceDirect, Springer, Emerald, and two prominent Iranian scholarly platforms (SID and Civilica). The search strategy employed key terms such as "supply chain implementation barriers," "circular supply chain implementation barriers," and "circular economy implementation barriers."

The inclusion criteria stipulated that selected publications must be published in 2018 or later, explicitly address CSC implementation, examine associated challenges, barriers, or pressures, and be English-language publications. This temporal focus reflects the emergence of 'circular supply chain' as a distinct research domain around 2017, with subsequent studies (post-2018) increasingly concentrating on implementation challenges and solutions. Figure 2 presents the article selection process following PRISMA guidelines, culminating in the inclusion of 30 relevant publications. For methodological rigor, the study engaged both industrial practitioners and academic specialists to assess the questionnaire's content validity. The expert panel was carefully selected to include professionals with demonstrated expertise in food industry operations, thereby ensuring the reliability of validation outcomes. Components failing to meet the minimum content validity ratio threshold (CVR < 0.33) were systematically excluded from further analysis. Moreover, Cohen's Kappa reliability coefficient was employed as the classification criterion for barrier categorization. This statistical measure quantifies agreement between two expert evaluators while accounting for the probability of expected agreement. The coefficient ranges from 0 to 1, where 0 denotes complete discordance between raters and 1 represents perfect concordance. Following established psychometric standards, a Kappa value exceeding 0.9 indicates near-perfect agreement, while values above 0.6 are generally considered acceptable for research purposes. In the present study, the computed Cohen's Kappa coefficient of 0.8 demonstrated substantial reliability. Based on this agreement, the identified barriers were systematically classified into five distinct dimensions: Economic, Infrastructure, Network, Managerial/Organizational, and Supply-related barriers. The complete validation results are detailed in Table 3.

The fuzzy SWARA questionnaire was developed, incorporating the five dimensions and sixteen barriers identified in previous research phases. Following refinement, the questionnaire was distributed to experts for evaluation. The data collection process comprised two distinct phases: In the initial phase, experts prioritized the identified barriers according to their perceived importance, assigning the highest priority to the most significant barrier. The subsequent phase required experts to conduct a comparative assessment of barriers, evaluating the relative importance of each barrier in relation to its predecessor within the established hierarchy. The final (global) weights for each barrier were calculated by multiplying its local relative weight by the corresponding dimensional weight. Table 4 presents the derived weights of all sub-criteria in descending order of importance. Following the barrier prioritization, big data-driven solutions were evaluated and ranked using the fuzzy Einstein

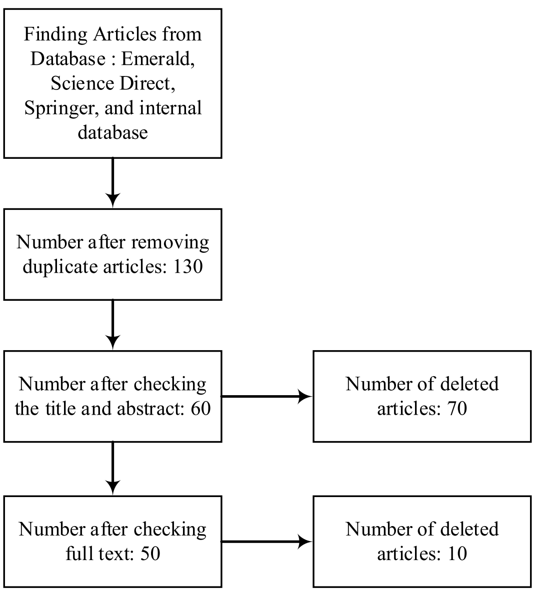

Fig. 2. Prisma diagram.

Criterion	Substandard	CVR	Source
Economic	Lack of government support for tax discount policies, incentives, subsidies, grants, etc. to promote circular business models	0.6	[41, 55, 69]
	Lack of financial benefits in the short term	0.6	[41, 69]
	High production costs	0.333333	[44, 47]
Infrastructure	Lack of necessary infrastructure for product tracking	0.6	[19, 69]
	Lack of necessary infrastructure for product recycling	0.333333	[33]
	Lack of organizational infrastructure to support CSC activities	0.333333	[9, 56, 76]
	Inadequacy of infrastructure facilities	0.333333	[5, 6, 15, 46, 54]
Network	Lack of knowledge, skills, awareness and participation of stakeholders	0.333333	[9, 44, 55, 56, 59, 76]
	Lack of cooperation/support from supply chain actors	0.333333	[19, 21]
Managerial/organizational	Lack of guidelines for end-of-life product management	0.333333	[49]
	Lack of senior management support and commitment to adopt circular procedures	0.333333	[29, 47, 55, 68, 78]
	Failure to accept new business models	0.333333	[2, 44, 67]
	Lack of performance evaluation system	0.733333	[26, 56]
	Lack of vision and strategic plan for the implementation of CSC activities	0.6	[44, 59, 78]
Supply	Lack of availability of recyclable materials	0.333333	[2, 22, 56]
	Low quality and quantity of returned products	0.333333	[2, 41, 46, 56]

Table 3. Barriers of circular supply chain in food industry.

Dimension	Dimension weight	Barrier	Fuzzy local weight	weight
Economic	(0.003, 0.005, 0.013)	Lack of government support for tax discount policies, incentives, subsidies, grants, etc. to promote circular business models	(0.714,0.800,0.837)	0.00549
		Lack of financial benefits in the short term	(0.119,0.171,0.279)	0.00133
		High production costs	(0.020,0.029,0.056)	0.00024
Infrastructure	(0.751, 0.822, 0.833)	Lack of organizational infrastructure to support CSC activities	(0.741, 0.817, 0.834)	**<u>0.6395</u>**
		Lack of necessary infrastructure for product tracking	(0.123,0.149 ,0.208)	**<u>0.12847</u>**
		Lack of necessary infrastructure for product recycling	(0.021,0.030,0.069)	0.02102
		Inadequacy of infrastructure facilities	(0.003,0.005,0.014)	0.00412
Network	(0.001, 0.001, 0.003)	Lack of knowledge, skills, awareness and participation of stakeholders	(0.750,0.813,0.857)	0.00127
		Lack of cooperation/support from supply chain actors	(0.125,0.188,0.286)	0.00031
Organizational management	(0.125, 0.147, 0.208)	Lack of vision and strategic plan for the implementation of CSC activities	(0.679,0.801,0.833)	**<u>0.12350</u>**
		Failure to accept new business models	()0.113,0.160,0.278	0.02942
		Lack of guidelines for end-of-life product management	(0.019,0.032,0.093)	0.0076
		Lack of senior management support and commitment to adopt circular procedures	(0.003,0.005,0.019)	0.00144
		Lack of performance evaluation system	(0.001,0.001,0.005)	0.00032
Supply	(0.021, 0.025, 0.052)	Lack of availability of recyclable materials	(0.833,0.857,0.857)	0.02781
		Low quality and quantity of returned products	(0.139,0.143,0.171)	0.00494

Table 4. The final weight of circular supply chain barriers. Bold and underline weights show the highest priority for evaluating barriers.

WASPAS methodology. In this stage, expert assessments focused on determining the efficacy of each proposed big data solution in addressing the identified CSC implementation barriers. The comparative ranking of these solutions relative to CSC barriers is presented in Table 5, with the complete calculation methodology detailed in Online Appendix C.

Discussion

The conceptual model developed in this study emerged from a comprehensive literature review and rigorous expert validation through questionnaire administration, followed by content validity assessment. The finalized framework comprises five distinct dimensions and sixteen identified barriers. Application of the fuzzy SWARA methodology revealed that the two most substantial barriers, both within the infrastructure dimension, collectively represent approximately 75% of the total weighting. This finding strongly suggests that infrastructure deficiencies constitute the principal impediment to circular supply chain (CSC) implementation. Theoretical contributions of this research demonstrate notable alignment with Khan and Ali's [40] findings in the pharmaceutical sector, particularly regarding the pivotal role of reverse logistics infrastructure. Additionally, organizational and managerial factors account for over 12% of total barrier weights, with the lack of strategic

Solutions	WSI	WPI	Di	Ranking
Social networks analysis	(0.58, 0.86, 1.02)	(11.43, 13.24, 15.89)	7.11	1
Optimization models	(0.61, 0.88, 1.03)	(11.36, 12.96, 15.30)	6.97	2
Cloud computing	(0.56, 0.81, 0.96)	(11.30, 12.96, 15.42)	6.95	3
Data mining	(0.60, 0.89, 1.06)	(11.22, 12.89, 15.33)	6.94	4
Artificial neural network	(0.61, 0.88, 1.05)	(11.09, 12.68, 15.04)	6.83	5
Machine learning	(0.58, 0.85, 1.01)	(10.96, 12.56, 14.86)	6.75	6
Statistical techniques	(0.54, 0.81, 0.95)	(10.69, 12.29, 14.48)	6.59	7

Table 5. fuzzy Einstein WASPAS method results.

vision for CSC implementation emerging as a prominent challenge. This observation corroborates Lahane and Kant's [49] identification of managerial and organizational support as critical CSC adoption barriers. However, comparative analysis reveals industry-specific variations in barrier significance. While Kazancoglu et al. [38] identified economic factors as predominant in dairy supply chains, followed by technological and environmental considerations, Khandelwal et al. [41] emphasized regulatory shortcomings in plastic industry contexts. These disparities underscore the contextual nature of CSC implementation challenges across industries and geographical regions. Methodologically, this study introduces innovation through application of the fuzzy Einstein-WASPAS hybrid technique for big data solution prioritization. This novel approach, combining WASPAS methodology with Einstein's fuzzy set theory, successfully addressed the inherent complexities and uncertainties of big data analytics. As anticipated, the analysis revealed minimal variation in solution effectiveness, attributable to substantial overlaps in big data techniques.

This study's findings yield significant managerial implications for implementing CSC practices. The developed framework serves as a strategic guide for organizational leaders seeking to overcome implementation barriers and realize CSC benefits. The study identifies inadequate organizational infrastructure as a critical barrier to CSC adoption. We recommend substantial investment in integrated management systems and advanced technologies to optimize information and material flows, enhance interdepartmental coordination, enable precise product tracking throughout lifecycle stages, minimize waste generation, and improve sustainability performance. Such infrastructure improvements yield dual benefits of enhanced operational efficiency and facilitated recycling/reuse processes.

To address the identified lack of managerial and organizational vision for circular activities, we propose establishment of a dedicated CSC task force, systematic evaluation of current organizational perspectives, and development of comprehensive strategic and operational plans incorporating clear circularity objectives, performance standards, and resource allocation frameworks.

The study advocates for SNA as a transformative tool for CSC optimization. SNA enables mapping of communication patterns across supply chain actors, identification of network strengths and vulnerabilities, enhancing the information flow velocity, and reduction of operational delays and costs. When combined with infrastructure improvements and strategic planning, these advanced analytical techniques can significantly elevate CSC performance and sustainability outcomes. Accordingly, Food industries should consider immediate infrastructure assessment and upgrading, development of formalized CSC implementation strategies, adoption of network analysis tools, and Continuous performance monitoring. This integrated approach addresses both technical and organizational dimensions of CSC implementation, providing a robust pathway toward CSC transformation.

Conclusion

This study employed MCDM techniques to prioritize big data-based solutions for addressing food supply chain implementation barriers. The research utilized two advanced fuzzy methods. Fuzzy SWARA enabled the determination of criterion weights through fuzzy quantification of expert judgments, subsequently transformed into crisp numerical values. Fuzzy Einstein WASPAS combined fuzzy logic with both additive and multiplicative models to evaluate alternatives against multiple criteria, yielding more robust results under conditions of uncertainty. These techniques proved particularly valuable for managing imprecise information and incorporating expert knowledge, thereby enhancing decision-making quality in ambiguous operational environments. The research faced significant constraints regarding its statistical population. The nascent stage of CE adoption in Iran's food industry resulted in a limited pool of qualified experts, restricting the study to a relatively small expert panel. This limitation affects the generalizability of findings, which would ideally require broader industry participation. Practical constraints, including budgetary limitations and time considerations, necessitated concentration on food industries in southern Iran, potentially limiting the nationwide applicability of results. The current undeveloped state of alternative fuzzy Einstein MCDM techniques precluded comparative analysis and sensitivity testing of results. These limitations naturally affect the scope and external validity of the findings. Building upon the findings of this study, the following research directions are proposed to advance scholarly understanding and practical implementation of CSCs. Future studies should undertake systematic comparisons between the Fuzzy Einstein approach and alternative analytical techniques. Given the current research gap in developing economies, we recommend expanding empirical investigations across multiple industrial sectors and developing of localized frameworks and solutions. This would significantly contribute to the global knowledge base on CSC adoption. Lastly, since this research has demonstrated that SNA is one of the most effective big data-based solutions for overcoming the challenges of CSC implementation, it is recommended to conduct a case study on the effects and outcomes of this method. Such a study can contribute to a deeper understanding of the application of social network analysis in enhancing the performance and sustainability of the CSC, as well as identifying related opportunities and threats.

Data availability
The research data can be accessed by contacting the corresponding author of the research.

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Author contributions
Rohham Farzadnia conducted the review, performed the fuzzy analysis and wrote the initial draft and revision of the manuscript. Payam Shojaei supervised the whole process of conducting the research and conceptualization of the model and final revision. Seyed Hadi Mirghaderi collaborated in the data collection and collaborated in writing of the manuscript.

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